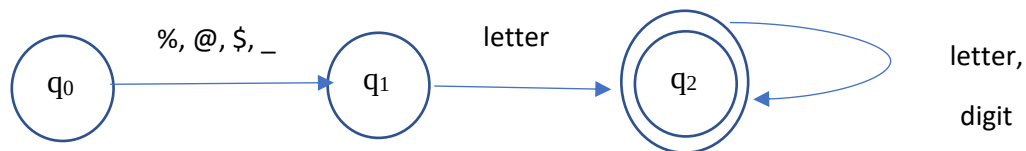


CEN202 Programming Languages Final Homework

SOLUTIONS SET

1. In this question you were required to write your own opinions and justify your opinions with some discussions by giving examples from some programming languages.
2. State diagram for the described variable name is shown below:



The BNF rules are:

```
<variable> → <symbol><letter> | <symbol><letter><letter_digit>
<symbol> → % | @ | $ | _
<letter> → a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | A | B |
C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z
<letter_digit> → <letter><letter_digit> | <digit><letter_digit> | ε
<digit> → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
```

(When writing the BNF or EBNF rules you have to be careful that symbols starts with < and ends with > symbols are non-terminal symbols, other are terminals. If you don't be careful about this you loose some points. You should be very careful about the notations used.)

Leftmost derivation for @x1YZ

```
<variable> => <symbol><letter><letter_digit>
=> @<letter><letter_digit>
=> @x<letter_digit>
=> @x<digit><letter_digit>
=> @x1<letter_digit>
=> @x1<letter><letter_digit>
=> @x1Y<letter_digit>
=> @x1Y<letter><letter_digit>
=> @x1YZ<letter_digit>
=> @x1YZ
```

(In the leftmost derivation, you have to apply each rule one by one. If you apply multiple rules in one step, you loose some points)

3. The given code finds and displays the row number of the first row of an n by n matrix such that all elements in the row are equal to 0.

We can write the same code without goto as follows **(of course there are other ways, but if your code finds the index of the row that has only one 0 element, you don't get any point):**

```
int flag;
for (i=0; i < n; i++){
    flag = 0;
    for (j=0; j < n; j++){
        if (x[i][j]!=0)
            flag = 1;
    }
    if (flag == 0){
        printf("First row that satisfies the condition is %d\n",
i);
        break;
    }
}
```

Although goto command makes program unreadable, in this question, using goto improves readability. Because when a non-zero element is observed in the row, the code rejects this row by using goto command. In my code, on the other hand, I used a flag variable, and it can't describe the meaning as that of goto.

4. Using the structures parent (X, Y), male (X), and female (X), the structures that define father (X, Y), mother (X, Y), brother (X, Y), sister (X, Y), grandparent (X, Y), and grandmother (X, Y) can be written as follows **(if you use other structures like sibling etc.; if you don't be careful about the usage of capital and small case letters you loose some points. You have to be careful that capital letter means variable, small case letters denote constants).**

```
father(X,Y) :- parent(X,Y), male(X).
mother(X,Y) :- parent(X,Y), female(X).
brother(X,Y):- parent(Z,X), parent(Z,Y), male(X).
sister(X,Y):- parent(Z,X), parent(Z,Y), female(X).
grandparent(X,Y):- parent(X,Z), parent(Z,Y).
grandmother(X,Y):- mother(X,Z), parent(Z,Y).
```